

Inflatable Magnetic Soft Tactile Sensor

Internship Project S26

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Project Overview

Soft inflatable sensor that converts physical contact into measurable force and contact information

Research Motivation

Soft robotic grippers:
require tactile feedback to understand how and where they contact an object

Conventional rigid sensors:
difficult to integrate into deformable surfaces

Develop a **pneumatically actuated soft sensor** that can function as a **soft gripper** while sensing **three-axis contact forces** and **contact orientation**

Research Contribution

Inflatable tactile sensor that preserves force and contact-state estimation as its **geometry and compliance change with internal air pressure**

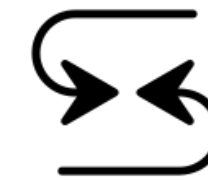
Research Goal

inflatable tactile sensor capable of measuring contact state across varying internal air pressures :

3-axis Force



Contact Orientation

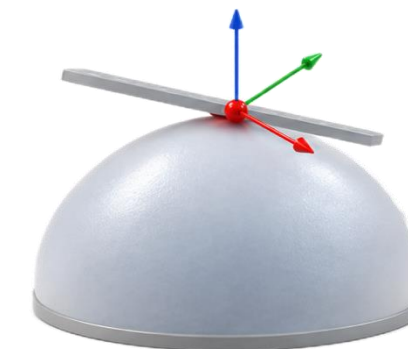
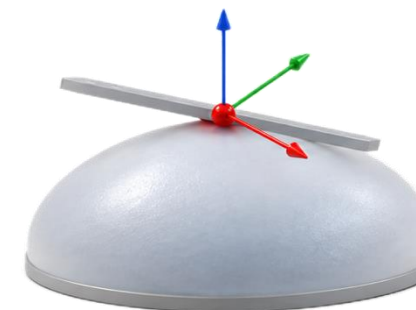


Contact Location



Low Pressure

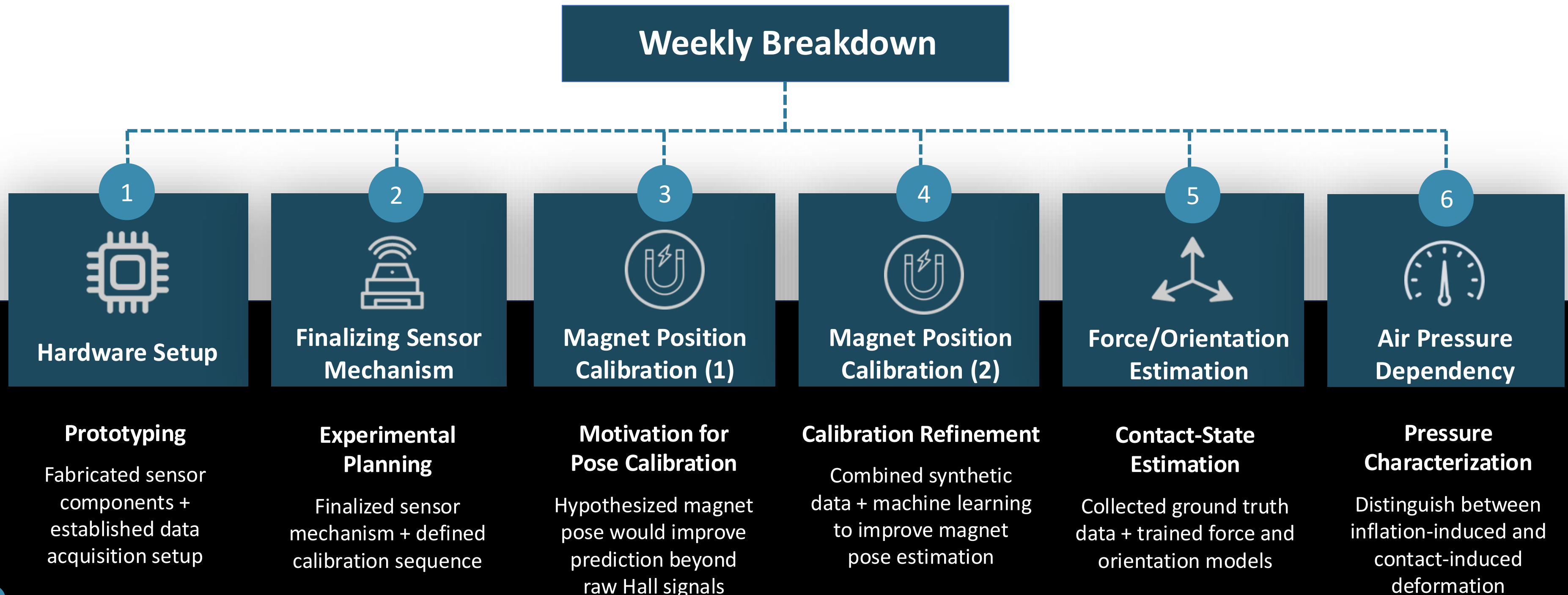
High Pressure



Measure contact reliably even when internal air pressure changes the shape of the soft sensor

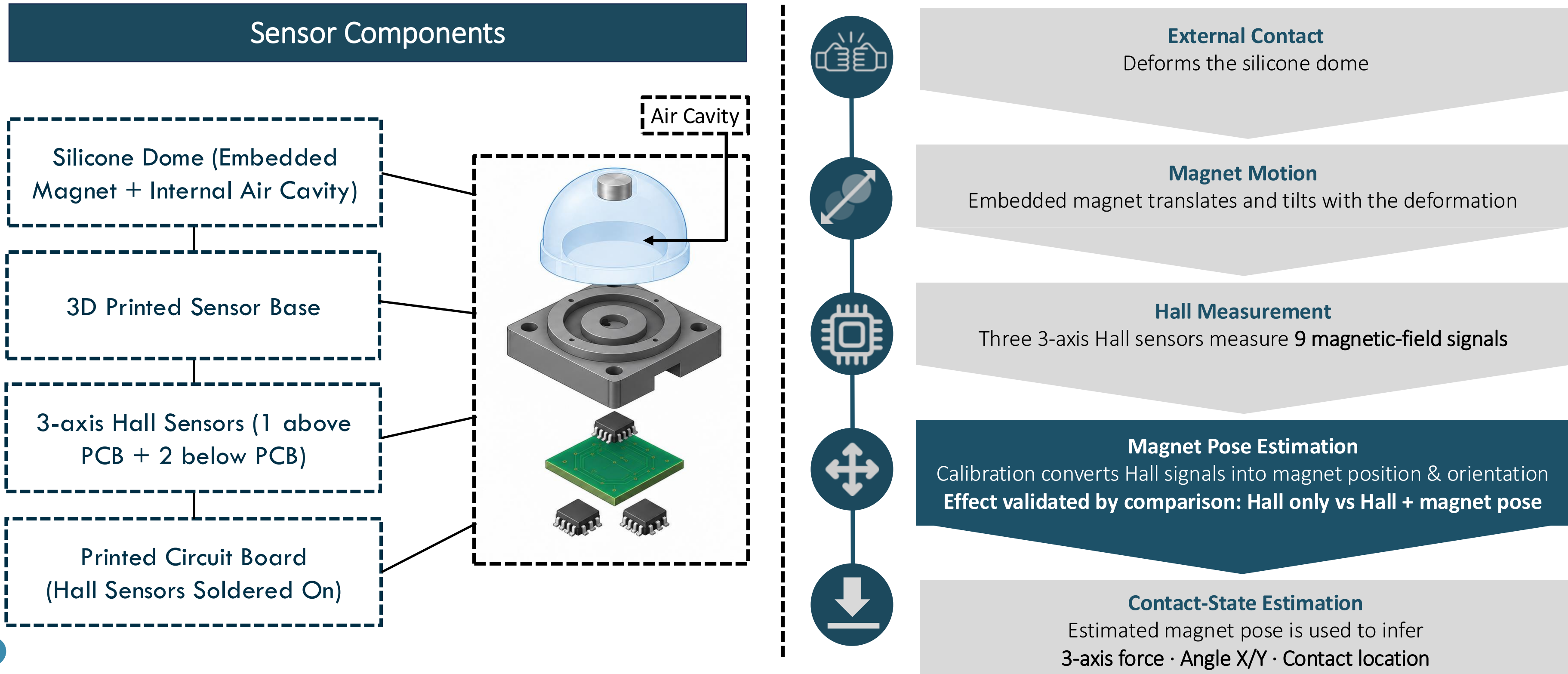
Progress Outline

Weekly deliverables and result sequence



Sensing Principle

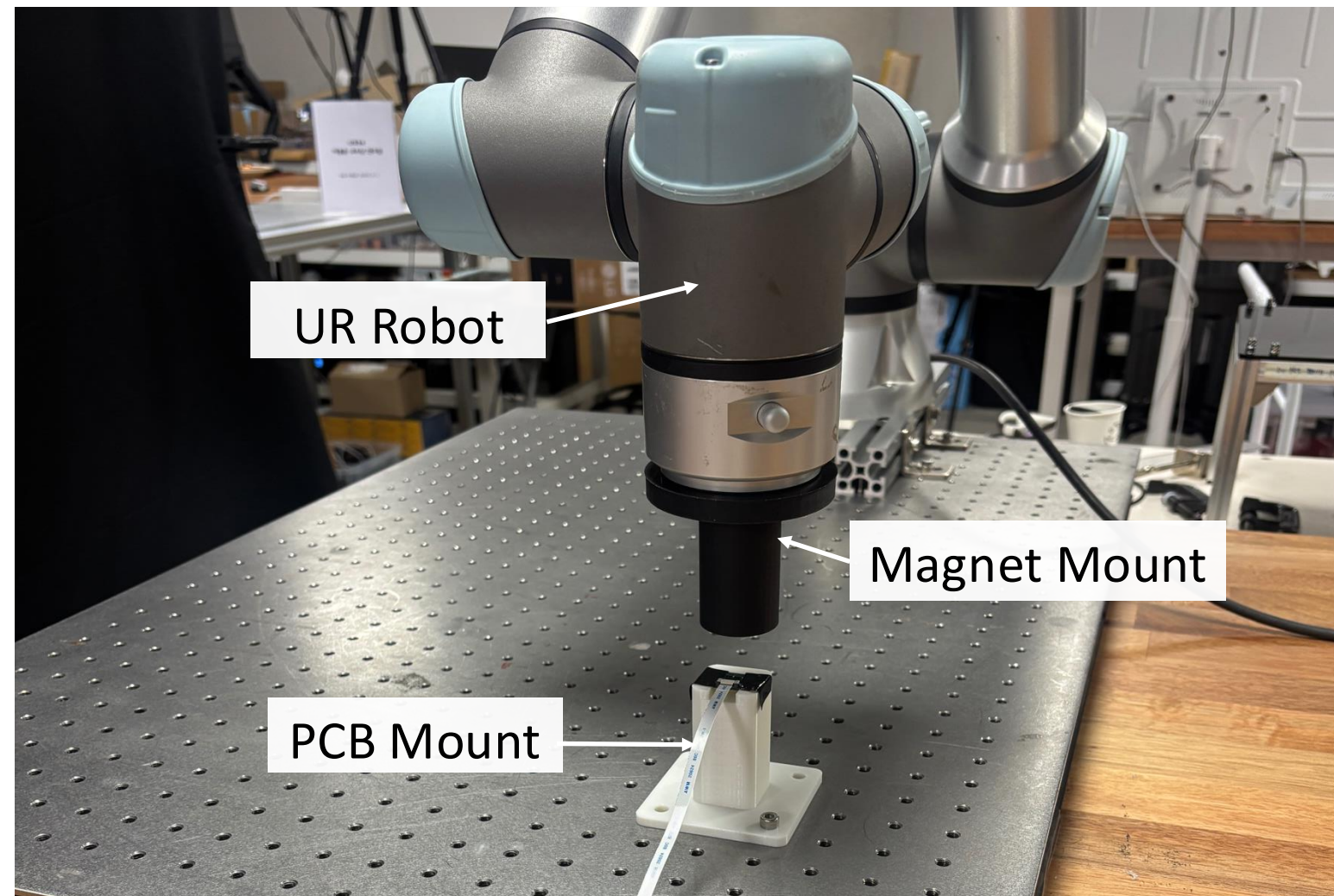
How sensor deformation is converted into contact information



Magnet Position Calibration (1)

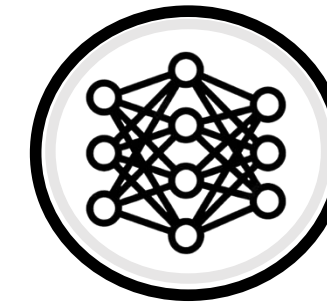
Mapping hall sensor signals to magnet position and orientation

Data Collection Setup



- UR robot moves the magnet through known positions and tilts
- 9 Hall signals are recorded at each commanded magnet pose

Calibration Methodology



Challenge: Limited Real Ground-Truth Data

Dense sampling across the full magnet-pose range requires hours of robot data collection



Solution: Physics Based Synthetic Data

Sample additional magnet poses across workspace and simulate their corresponding Hall signals



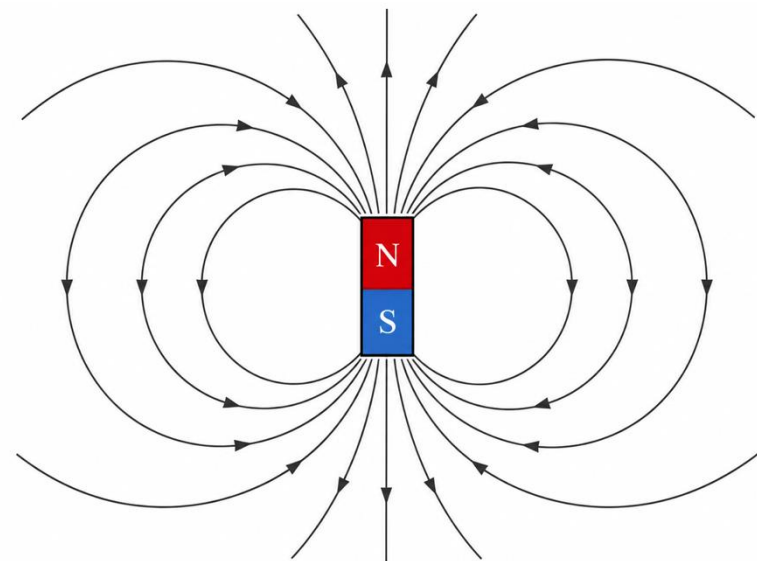
Model Training: Synthetic Pretraining + Real Data Fine-Tuning

Train the pose-estimation model on synthetic data, then fine-tune with limited real measurements

Magnet Position Calibration (2)

Combining physics simulation and machine learning for 5-DOF magnet pose estimation

Physics Model: Synthetic Pose Generation



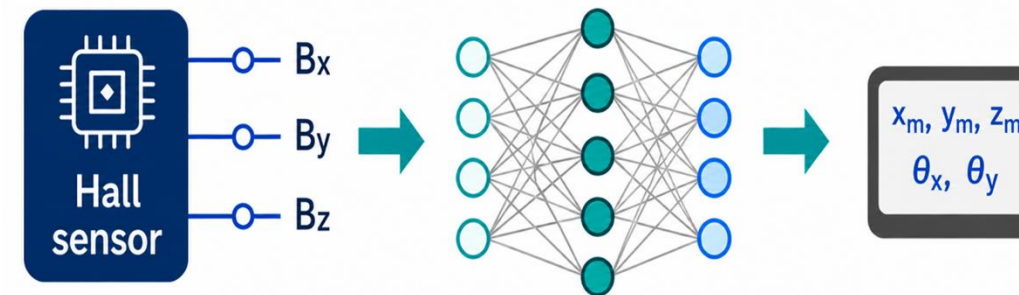
Sample Magnet Poses

Generate poses across the workspace
 $q = [x, y, z, \theta_x, \theta_y]$

Simulate Hall Signals

Calculate magnetic field seen by all 3 Hall sensors → 9 synthetic Hall signals per pose

Machine Learning: Multi Layer Perceptron (MLP)



Synthetic Pretraining

Train the MLP on simulated Hall signals → known magnet pose

Real Data Fine-tuning

Fine-tune using limited UR-collected data to adapt the model to the physical sensor

Calibration Issues + Refinement

Translation – Tilt Ambiguity

Translation and tilt created similar Hall patterns
 $x \leftrightarrow \theta_y, y \leftrightarrow \theta_x$

Fix: Added cross-sensor difference features to better separate coupled motions

Near-field Error

Prediction error increased when the magnet was closest to Hall sensors

Fix: Introduced proximity-aware calibration for the near-field region

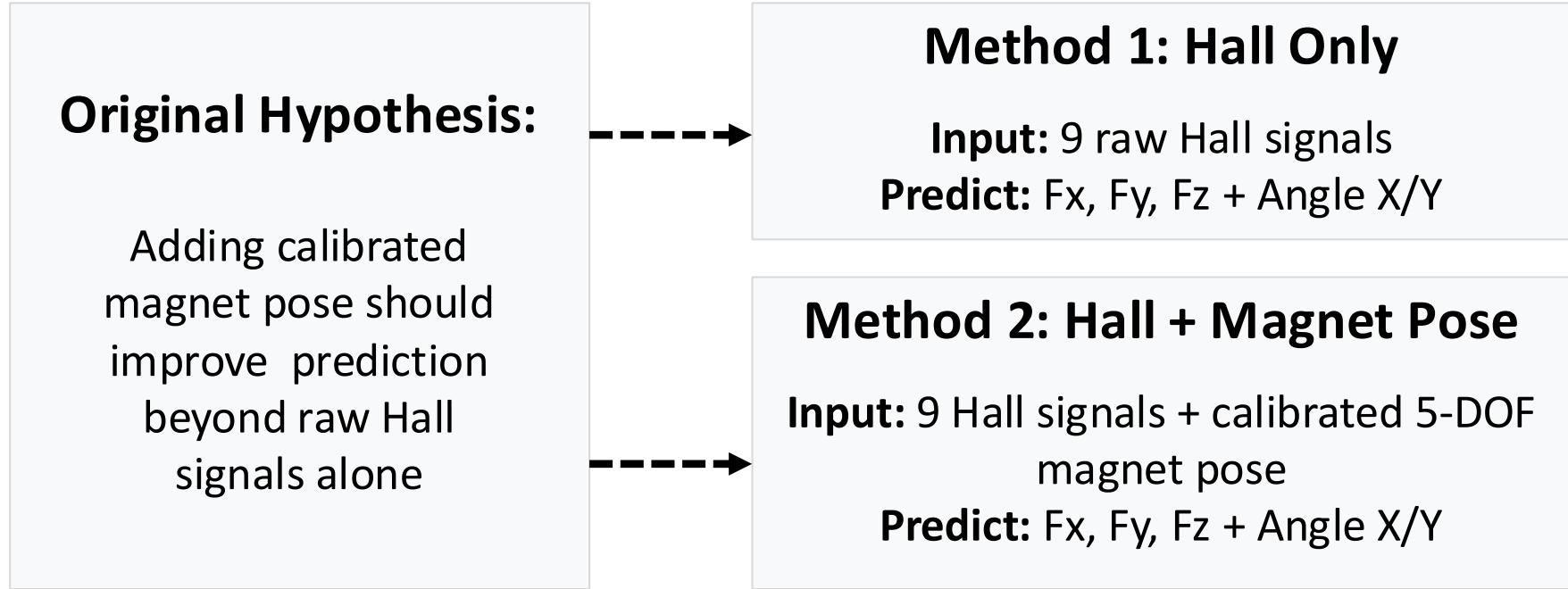
Uneven Pose Coverage

Training samples were concentrated near the center of the workspace

Fix: Expanded pose ranges and balanced synthetic/real sampling across the workspace

Testing if magnet-pose information improves force and contact-orientation estimation

Input Comparison: Does Magnet Pose Improve Prediction?



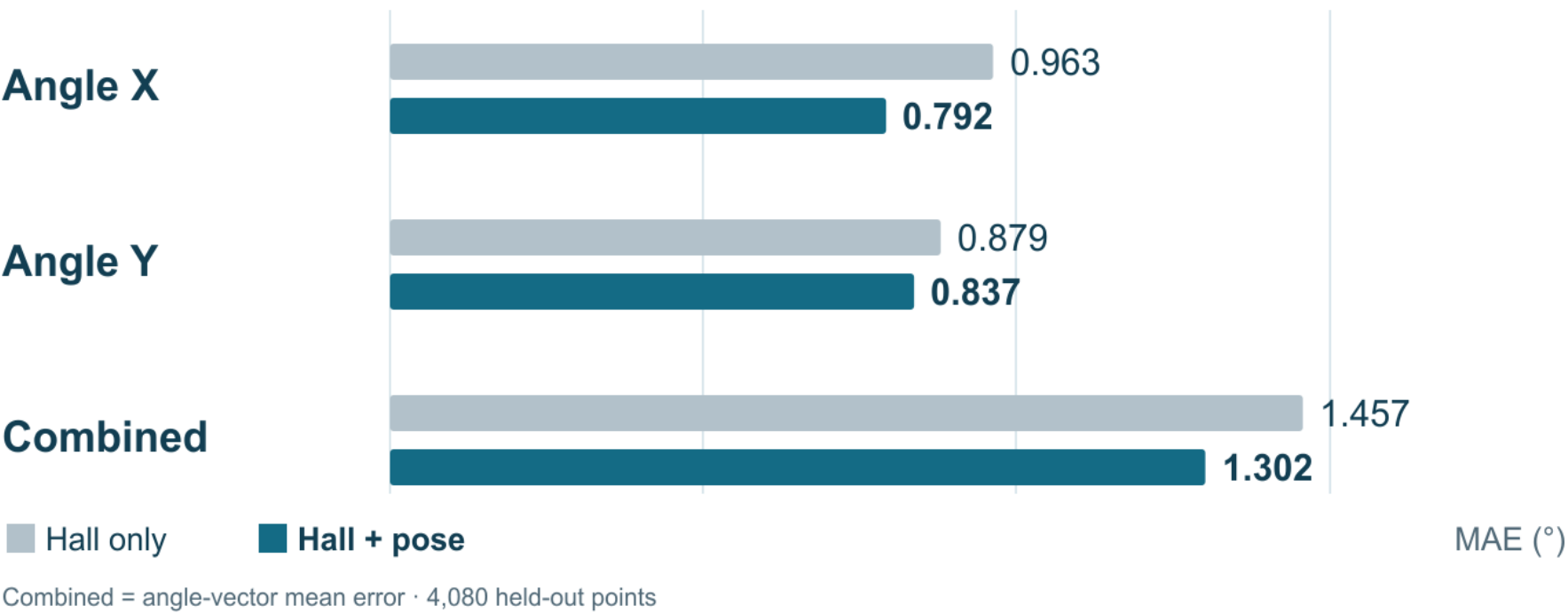
Result Interpretation

- Orientation improved:** Adding magnet pose reduced combined orientation MAE by ~11%
- Force improvement was target-dependent:** Largest gains occurred for **F_y** and **shear**, while F_x and F_z changed little

Takeaway: Magnet pose adds useful physical information, particularly for **orientation and lateral-force estimation**

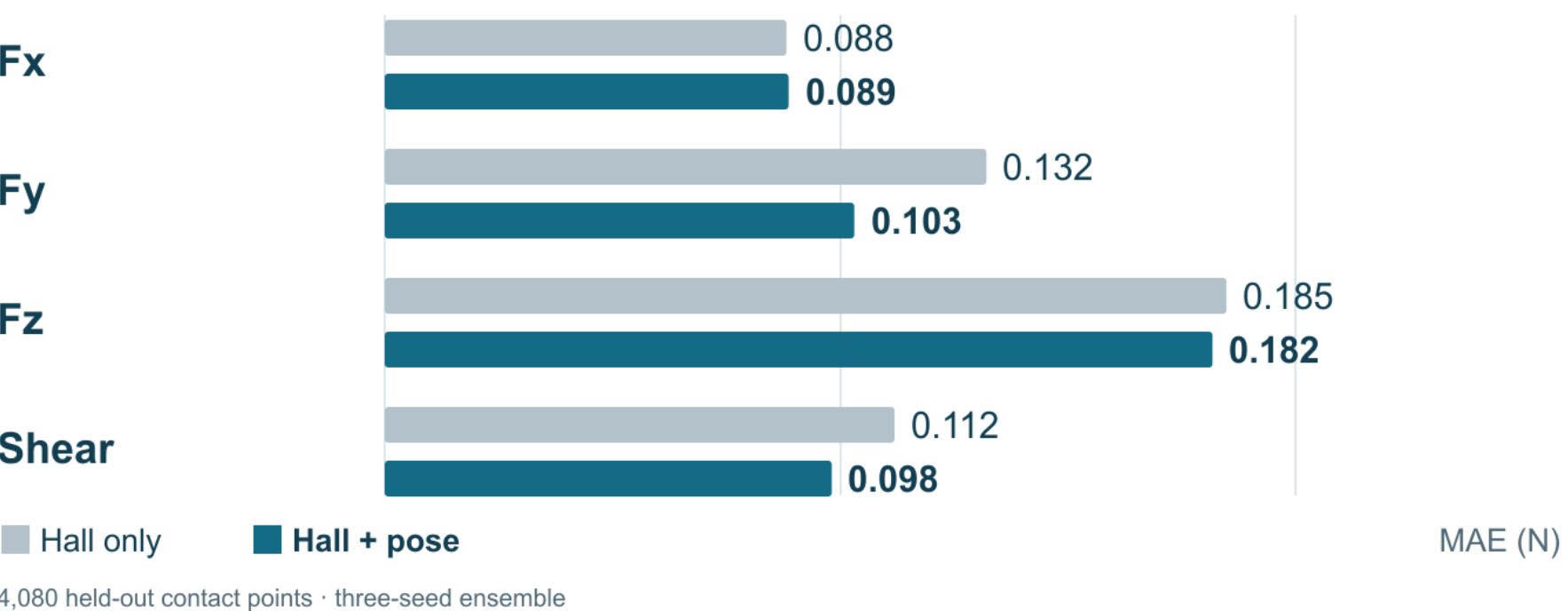
Contact orientation error

LOWER IS BETTER



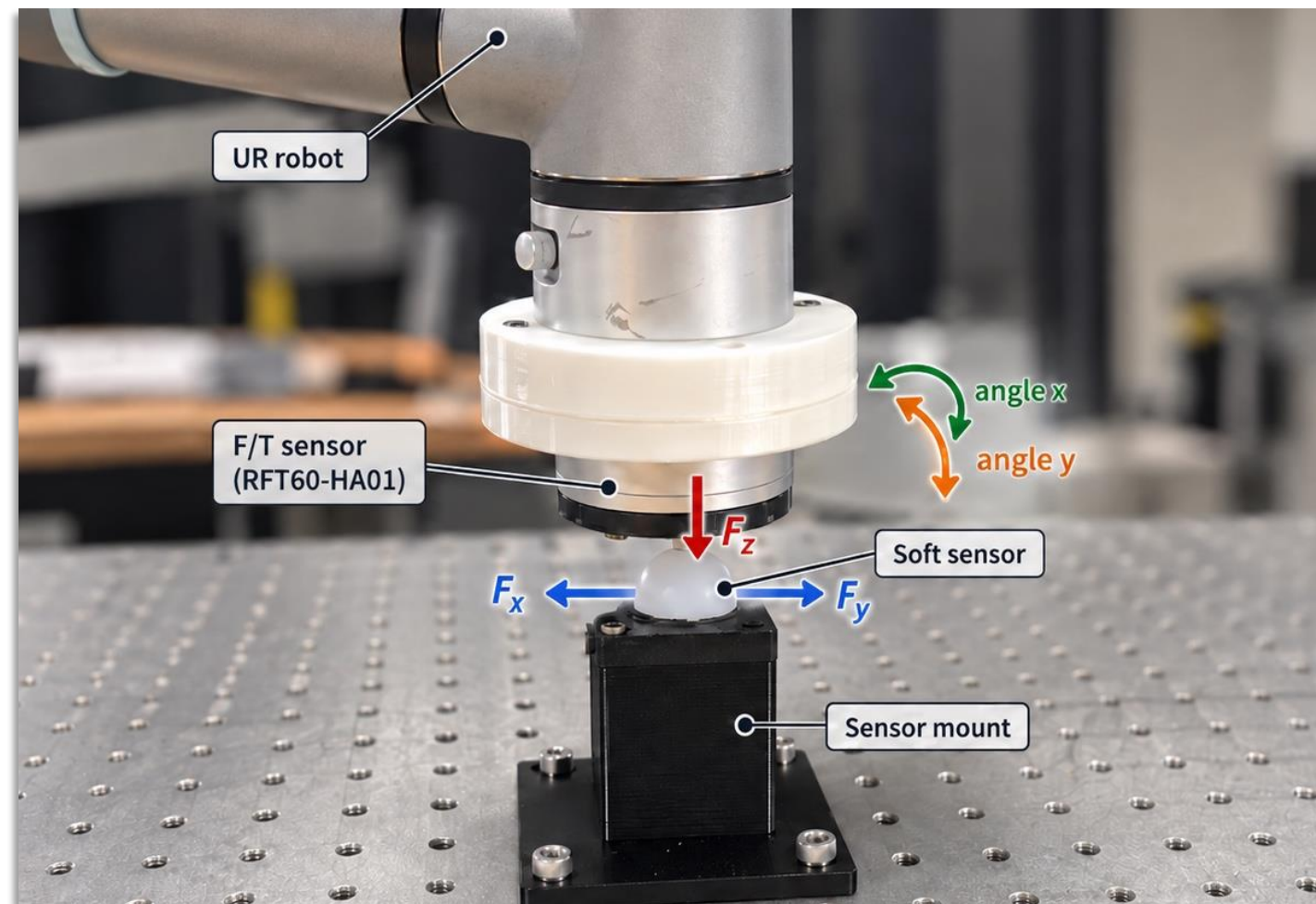
Force prediction error

LOWER IS BETTER



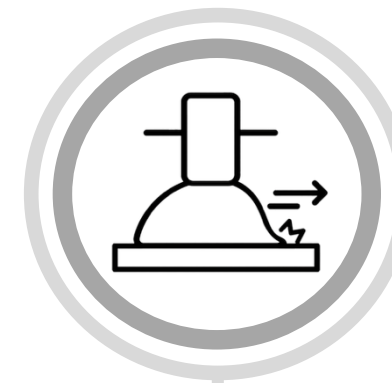
Force & Sensor Orientation Estimation

Ground-truth data collection and key sources of prediction error



Ground Truth Data Collection

- UR robot applied controlled normal, shear, and tilt motions
- RFT sensor measured F_x , F_y , F_z while Hall signals were recorded simultaneously
- Robot pose provided ground-truth Angle X / Angle Y



Slippage

Large shear displacement caused contact surface to slip, making force measurements unreliable

Solution

Restrict shear range to ± 2 mm
Keep only no-slip region for training



Hysteresis

Silicone hysteresis caused different sensor responses during loading and unloading at same position

Solution

Temporal Modeling
Use recent signal history to capture motion-dependent behavior



Large Error Regions

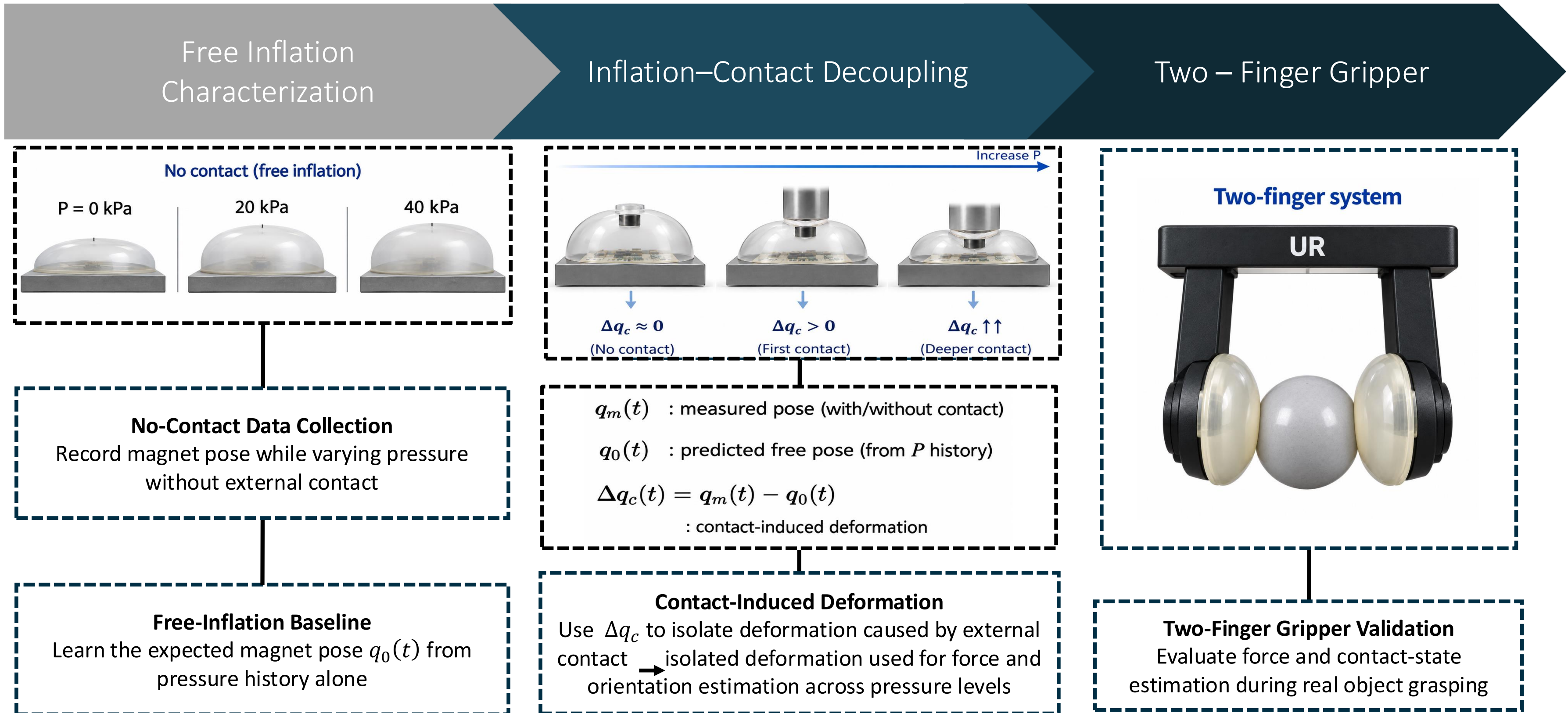
Prediction error was concentrated in specific, underrepresented regions of contact workspace

Solution

Targeted Data Collection (In Progress)
Collect additional samples in high-error regions

Air Pressure Dependency (Next Steps)

Extending tactile sensing across varying air pressures

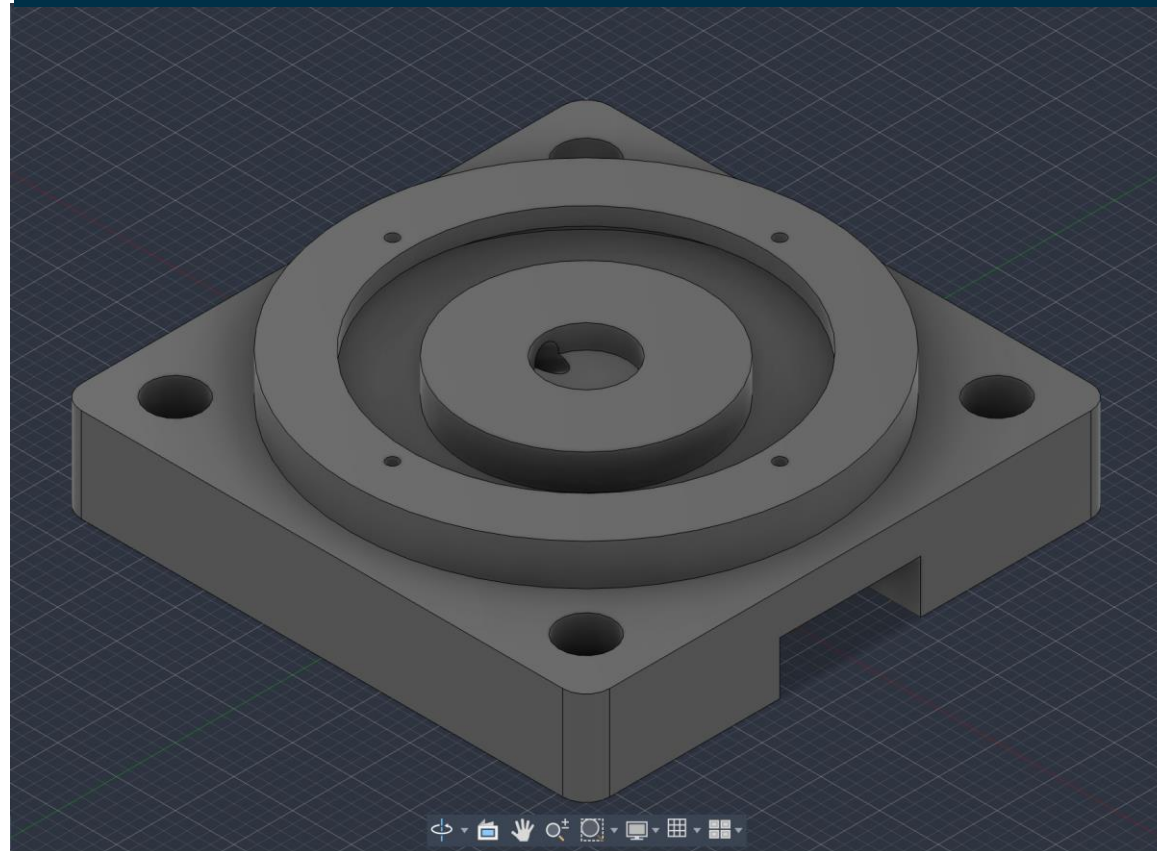


Appendix

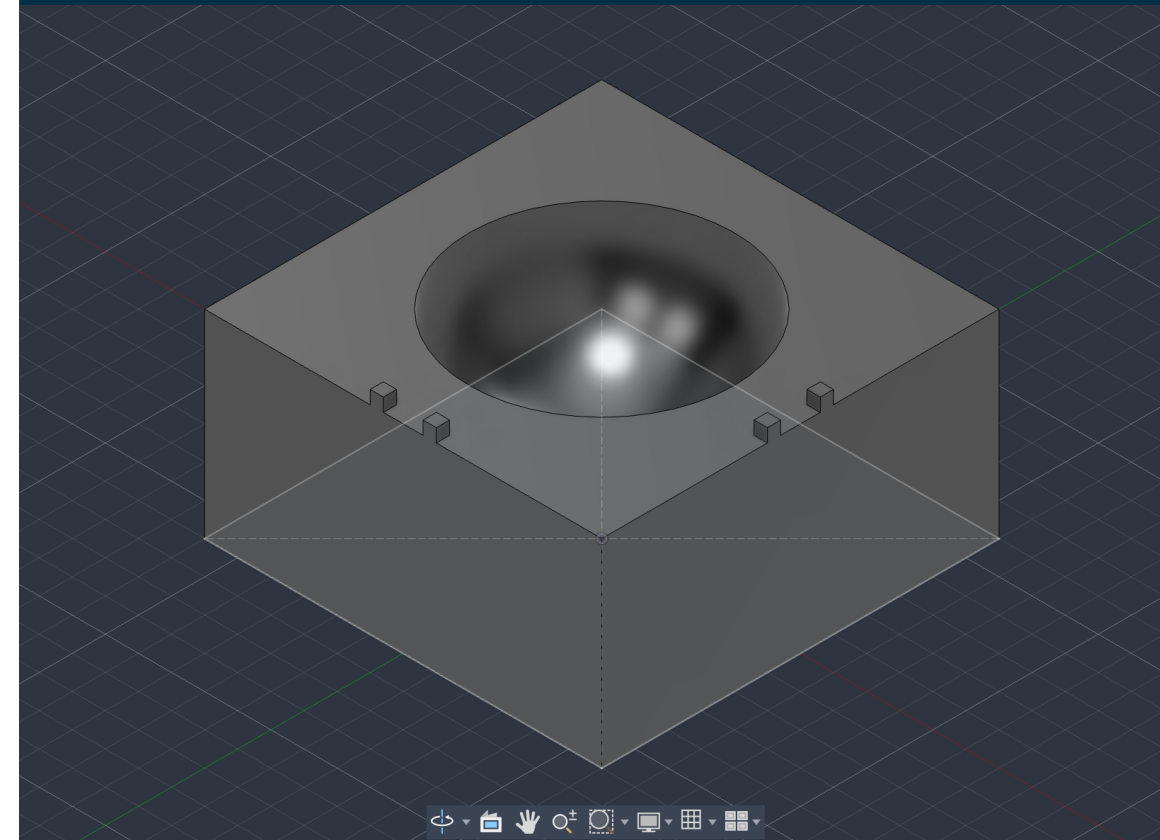
CAD files generated for the sensor

Made with Autodesk Fusion software / 3D printed using Formlabs

Sensor Base



Silicon Mold



Mold Presser

